

KEUN-SOO HEO (허근수)



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RESEARCH INTERESTS

- **Trustworthy Vision–Language Models**
 - Self-correction and verification-driven learning to reduce hallucinations in multimodal generation
- **Data-Efficient Learning under Limited Data and Annotation**
 - Meta-learning and semi-supervised approaches for effective learning from sparsely labeled data
- **Domain-Adaptive Multimodal Representation Learning**
 - Learning multimodal representations robust to domain shift across heterogeneous datasets

EDUCATION & RESEARCH EXPERIENCE

Korea University

Integrated M.S.-Ph.D. in Department of Artificial Intelligence

- Medical Artificial Intelligence Lab. (Advisor : Prof. Tae-Eui Kam)
- GPA: 4.31/4.5

Seoul, South Korea

Mar. 2020 – Aug. 2026 (Expected)

Jan. 2020 – Present

The Catholic University of Korea

B.S. in Department of Information, Communication and Electronic Engineering

- Image Signal Processing Lab. (Advisor : Prof. Changwoo Lee)
- GPA: 4.33/4.5

Bucheon, Gyeonggi-do, South Korea

Mar. 2014 – Feb. 2020

Dec. 2017 – Jan. 2020

* Served in the Republic of Korea Army as part of mandatory military duty (2015–2017)

TEACHING EXPERIENCE

- **AI Training Program for LG CNS, Korea University**
 - Teaching Assistant, Data AI (Instructor: Prof. Sejun Park) Sep. 2023 – Nov. 2023
 - Teaching Assistant, Machine Learning (Instructor: Prof. Tae-Eui Kam) May 2023 – Jun. 2023

SCHOLARSHIPS

- **KOBRA Travel Award (KOBRA-TA)**, Korea Bio Research Association
 - Jun. 2025 — KRW 3,000,000
- **BK21 FOUR Graduate Research Scholarship**, Ministry of Education, Korea
 - Spring 2022, Fall 2021, Spring 2021
- **Graduate School Scholarship**, Korea University
 - Fall 2020

- **Research Assistantship**, Korea University
 - Spring 2021
- **Outstanding Academic Scholarship**, The Catholic University of Korea
 - Fall 2019, Spring 2019, Fall 2018, Spring 2018, Fall 2015
- **Presidential Entrance Scholarship**, The Catholic University of Korea
 - Spring 2014, Fall 2014

AWARDS

- MICCAI 2025 Oral Presentation & Highlight Paper, 2025
- First Place Graduate in Department (Valedictorian), The Catholic University of Korea, 2020
- Honorable Mention Award, Seoul App Development Competition, Seoul Metropolitan Government, 2019

PUBLICATIONS

Keun-Soo Heo, Ji-Wung Han, Soyeon Bak, Minjoo Lim, Bogyong Kang, Sang-Jun Park, Weili Lin, Han Zhang, Dinggang Shen, and Tae-Eui Kam, “Sparsely Labeled fMRI Data Denoising with Meta-Learning-Based Semi-Supervised Domain Adaptation,” *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025. (**Oral Presentation (Top 2%)**, Early Accepted (Top 9%))

Bogyong Kang, Sang-Jun Park, Minjoo Lim, Myeongkyun Kang, **Keun-Soo Heo**, Ji-Hye Oh, Hyun Jung Lee, and Tae-Eui Kam, “Pre-to-Post Operative MRI Generation with Retrieval based Visual In-Context Learning,” *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.

Sang-Jun Park, **Keun-Soo Heo**, Bogyong Kang, Minjoo Lim, and Tae-Eui Kam, “Group-wise Compression and Summarization via LLM-based Ensemble for Chest X-ray Report Generation,” *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2025. (**Oral Presentation**)

Sang-Jun Park*, **Keun-Soo Heo***, Dong-Hee Shin, Young-Han Son, Ji-Hye Oh, and Tae-Eui Kam, “DART: Disease-aware Image-Text Alignment and Self-correcting Re-alignment for Trustworthy Radiology Report Generation,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. (*: Equal contribution)

Bogyong Kang, Hyeonyeong Nam, Myeongkyun Kang, **Keun-Soo Heo**, Minjoo Lim, Ji-Hye Oh, and Tae-Eui Kam, “Target-Aware Cross-Modality Unsupervised Domain Adaptation for Vestibular Schwannoma and Cochlea Segmentation,” *Scientific Reports*, 2024.

Jun-Mo Kim, **Keun-Soo Heo**, Dong-Hee Shin, Hyeonyeong Nam, Dong-Ok Won, Ji-Hoon Jeong, and Tae-Eui Kam, “A Learnable Continuous Wavelet-based Multi-Branch Attentive Convolutional Neural Network for Spatio-Spectral-Temporal EEG Signal Decoding,” *Expert Systems With Applications*, 2024.

Sanghyeon Cho, Bogyong Kang, **Keun-Soo Heo**, Eunjung Jo, and Tae-Eui Kam, “Enhanced Structure Preservation and Multi-View Approach in Unsupervised Domain Adaptation for Optic Disc and Cup Segmentation,” *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024.

Minjoo Lim, **Keun-Soo Heo**, Jun-mo Kim, Bogyong Kang, Weili Lin, Han Zhang, Dinggang Shen, and Tae-Eui Kam, “A Unified Multi-Modality Fusion Framework for Deep Spatio-Temporal-Spectral Feature Learning in Resting-State fMRI Denoising,” *IEEE Journal of Biomedical and Health Informatics*, 2024.

Bogyong Kang, Hyeonyeong Nam, Ji-Wung Han, **Keun-Soo Heo**, and Tae-Eui Kam, “Multi-view Cross-Modality MR Image Translation for Vestibular Schwannoma and Cochlea Segmentation,” *MICCAI BrainLes Workshop*, 2022.

Keun-Soo Heo, Dong-Hee Shin, Sheng-Che Hung, Weili Lin, Han Zhang, Dinggang Shen, and Tae-Eui Kam, “Deep Attentive Spatio-Temporal Feature Learning for Automatic Resting-State fMRI Denoising,” *NeuroImage*, 2022.